

Mohith

mohith.work@outlook.com — github.com/mohithgundimeda — Portfolio — Blogs

EXPERIENCE

Freelance ML Engineer

LLM-based Draft Automation System — GitHub

2025 – Present

- Built a multilingual document automation system that converts Telugu instructions into structured XML documents and DOCX outputs using transformer-based LLM pipelines.
- Fine-tuned Qwen 2.5 7B with LoRA for controlled structured generation, improving output reliability and reducing hallucinations.
- Engineered a multi-stage NLP pipeline covering document classification, structured extraction, generation, and output validation.
- Built and deployed an end-to-end application for automated document workflows, structured form filling, and downloadable document generation.
- Developed prompt refinement and structured-generation strategies to improve task-specific output quality and consistency.

Freelance Full Stack Engineer

Shivira Travels — Website

2025 – Present

- Built and deployed a production full-stack travel platform supporting bookings, enquiries, and package discovery.
- Developed responsive React interfaces with dynamic routing, search, and interactive travel workflows.
- Engineered Node.js and Express backend services for authentication, APIs, administration, and email workflows.
- Managed production deployment, hosting, domain configuration, and ongoing maintenance for business operations.

AI Intern

Slash Mark — Certificate

Jan 2024 – Apr 2024

- Built and evaluated machine learning and NLP models for classification and text analysis across multiple datasets.
- Implemented CNN architectures and NLP preprocessing pipelines for model training and evaluation.

Data Science Intern

Solar Secure Solutions — Certificate

Oct 2023 – Dec 2023

- Performed statistical analysis and developed predictive models using Python for cost-related analysis.
- Built and evaluated regression models to identify patterns and generate cost predictions.

PROJECTS

Make Sense — Research Paper Explainer Pipeline — GitHub

- Built an end-to-end pipeline that transforms research paper PDFs into structured, beginner-friendly explainer PDFs using LLM-based content restructuring.
- Engineered a CPU-only document extraction pipeline using Marker and a local llama.cpp inference server, preserving formulas, tables, and figures without GPU infrastructure.
- Implemented a multi-provider LLM fallback architecture across OpenRouter and Gemini to handle rate limits, provider failures, and malformed responses.
- Designed a configuration-driven system using Pydantic settings and JSON/INI configuration to control model selection, retry policies, and processing parameters.

- Automated PDF generation using Pandoc and Tectonic, producing LaTeX-based documents with preserved mathematical notation and consistent styling.
- Built a Gradio interface for end-to-end pipeline execution with structured logging and error handling across processing stages.

Dynamodelx — Dynamic ML Modeling Framework — GitHub

- Built a PyTorch-based framework for dynamically constructing and training machine learning models for regression, classification, and uncertainty estimation.
- Engineered modular APIs for rapid experimentation across statistical and machine learning workflows.
- Implemented Bayesian Neural Networks using variational inference and ELBO optimization for probabilistic and uncertainty-aware modeling.
- Built reusable experimentation workflows for training, evaluation, and model comparison.
- Automated model training and evaluation pipelines for repeatable experimentation and validation.
- Implemented a unified framework supporting heteroscedastic regression and multi-class classification.

Baseline OS — Minimal Linux Runtime for Production ML — Google Drive

- Built a minimal Arch Linux runtime as an alternative to generic cloud-provider OS images, removing unnecessary system services and packages to minimize idle resource consumption.
- Configured the operating system, networking, package management, and initialization stack to create a lean workload-specific production environment.
- Optimized the runtime around a dedicated ML workload, allocating system resources primarily to model execution and inference.
- Containerized ML dependencies with minimal Docker environments to maintain reproducible builds and consistent deployment across hosts.

SKILLS

Programming: Python

Machine Learning: PyTorch, scikit-learn, CNNs, Bayesian Neural Networks, Statistical Modeling, Regression, Classification

NLP and LLMs: Transformers, LoRA, Fine-tuning, Tokenization, Structured Generation, Prompt Engineering, Text Classification

ML Workflows: Model Training, Evaluation Pipelines, Experimentation, Inference Pipelines, Docker, Linux

Data Science Libraries: NumPy, Pandas, Matplotlib

Web Technologies: React, Node.js, Express, Flask, Streamlit

Tools and Platforms: Git, LangChain, Docker, Hugging Face, Pydantic

EDUCATION

Sri Venkateswara College of Engineering

B.Tech in Artificial Intelligence and Machine Learning

Tirupati, Andhra Pradesh

2021 – 2025